

Weekly Progress Report

Wednesday, July 15, 2026

- Completed internship registration and attended the **NU Research Overview and Internship Guide** session.
- Attended research sessions on:
 - Building the Researcher's Mindset: From Inquiry to Innovation
 - AI Ethical Usage in Research
 - Technical Writing: A Walk Through to be an Academic Author
- Participated in the introductory meeting with the Center Director and research track professors.

Thursday, July 16, 2026

- Attended sessions covering research methodology, including academic literature search strategies, publication venue selection, and the use of research databases.
- Attended a technical session introducing chaotic systems, their classifications, and evaluation metrics.
- Completed **Task 1: Simulation and Sensitivity Analysis of Chaotic Systems**.

Sunday, July 19, 2026

- Completed a literature review of **8 research papers** on the application of **Machine Learning and Deep Learning in Chaos Theory for Time Series Forecasting**.
- Organized the reviewed papers into a structured Excel database, documenting publication details, journal ranking, abstracts, limitations, and personal comments on the papers.

Monday, July 20, 2026

- Completed **Task 3: Evaluating Chaotic Systems**.
- Reviewed additional literature on **Reservoir Computing** and **ECG signal denoising**, with particular focus on advanced filtering techniques.
- Delivered a technical presentation demonstrating understanding of Reservoir Computing concepts and ECG denoising approaches.

Tuesday, July 21, 2026

- Attended a progress review meeting with the research team.
- Participated in technical discussions on ECG signal characteristics, advanced preprocessing techniques, and neural network architectures for chaotic time-series forecasting.
- Designed and implemented a comprehensive preprocessing pipeline for the **MIT-BIH Arrhythmia Database**.

Wednesday, July 22, 2026

- Continued investigating the proposed architecture for chaotic ECG time-series forecasting
- Investigated appropriate **train/validation/test splitting strategies** for ECG time-series data, with particular focus on **inter-patient** versus **intra-patient** splitting methodologies and their impact on model generalization and fair evaluation.

Overall Progress

This week focused on strengthening both the theoretical and practical foundations of the research through structured training sessions, literature review, completion of assigned technical tasks, development of an ECG preprocessing pipeline, and continued investigation of an architecture for chaotic ECG time-series forecasting.